

COURSE TITLE : CONSTRUCTION & BUILDING SERVICES LAB
COURSE CODE : 6188
COURSE CATEGORY : A
PERIODS/WEEK : 3
PERIODS/SEMESTER : 45
CREDITS : 2

TIME SCHEDULE

MODULE	TOPIC	PERIODS
I	Test on building materials	21
II	Setting out	6
III	Plumbing	12
IV	Wiring	6
TOTAL		45

Course Objective :

Sl.	Sub	Student will be able to
1	1	Conduct different tests on building materials
	2	Conduct compression test on concrete
	3	Set out a simple building.
2	1	List tools and materials required for water supply and sanitation
	2	Understand simple electrical wiring used for domestic purpose.

After completing this course the student will be able to:

Module I: Test on building materials

- Determine the fineness of cement.
- Determine standard and normal consistency of cement.
- Determine initial and final setting time of cement.
- Conduct compression test on concrete and find the stress at first crack and ultimate stress.
- Conduct tension test on steel and find yield point stress, breaking stress, ultimate stress, % of elongation and % of reduction in cross-sectional area of MS bar.
- Draw inferences from test results on workability of cement concrete construction.
- Conduct compression test on brick and find the stress at first crack.
- Determine the fineness modulus of coarse aggregate and fine aggregate

Module II: Setting out of buildings.

- Set out a simple building on site

Module III: Plumbing.

- List out tools and materials required for plumbing.
- Install a single tap and bath room shower.
- Install water closet with different types of traps.

Module IV: Wiring.

- Do the wiring for two lamps in series and parallel connection.
- Fix a two way switch.

CONTENT DETAILS

Module I: Test on building materials

Test on fineness of cement - standard and normal consistency of cement - Determine initial and final setting time of cement - compression test on concrete and find the stress at first crack and ultimate stress. - tension test on steel and find yield point stress, breaking stress, ultimate stress, % of elongation and % of reduction in cross-sectional area of MS bar - test on workability of cement concrete construction - compression test on brick and find the stress at first crack - fineness modulus of coarse aggregate and fine aggregate

Module II Setting out of buildings.

Set out of a simple building on site

Module III: Plumbing.

Tools and materials required for plumbing – Installation of a single tap and bath room shower – Installation of a water closet with different types of traps.

Module IV: Wiring.

Wiring for two lamps in series and parallel connection – Fixing of a two way switch.